

16x16 点阵显示实验

1、实验目的：

- (1) 学习点阵显示字符的基本原理。
- (2) 掌握用数控分频控制速度，实现点阵扫描的基本方式。

2、实验原理：

8X8 点阵的实际外观图如图 1：一共 64 个发光点构成，上下两排引脚，每排 8 个，一共 16 个点阵侧面有文字的那面对准自己。

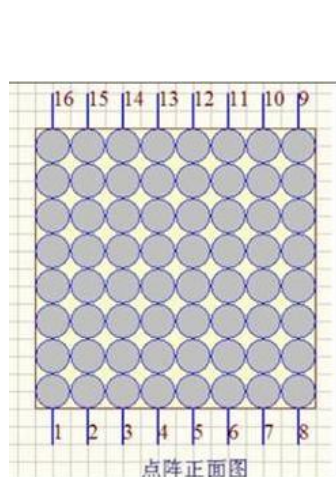


图 1 排列

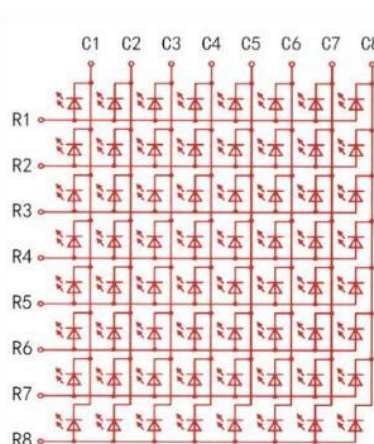


图 2 行列

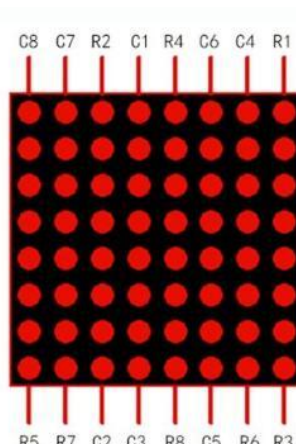


图 3 8*8 点阵实际引脚端

8X8 点阵的显示原理：

图 2，8X8 点阵共由 64 个发光二极管组成，R 为列，C 为行，且每个发光二极管是放置在行线和列线的交叉点上，当对应的某一行置 1 电平，某一列置 0 电平，则相应的二极管就亮；如要将第一个点点亮，则 R1 脚接高电平 C1 脚接低电平，则第一个点就亮了；如要将第一行点亮，则第 R1 脚要接高电平，而 C1~C8 这些引脚接低电平，那么第一行就会点亮；如要将第一列点亮。则第 R1 脚接低电平，而 C1~C8 接高电平，那么第一列就会点亮。图 3，实际点阵图，每个引脚对应的引脚名。

3、实验仿真：

根据描述语言进行仿真测试，分析波形。

4、管脚约束：

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set_pin_assignment { clk } { LOCATION = R7; IOSTANDARD = LVCMOS33; PULLTYPE =  
PULLUP; }  
set_pin_assignment { dataout[0] } { LOCATION = K12; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { dataout[10] } { LOCATION = G16; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { dataout[11] } { LOCATION = G14; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { dataout[12] } { LOCATION = K15; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { dataout[13] } { LOCATION = J16; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { dataout[14] } { LOCATION = H15; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { dataout[15] } { LOCATION = H16; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { dataout[1] } { LOCATION = L14; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { dataout[2] } { LOCATION = M14; IOSTANDARD = LVCMOS33;  
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set_pin_assignment { dataout[3] } { LOCATION = P11; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { dataout[4] } { LOCATION = L10; IOSTANDARD = LVCMOS33;  
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set_pin_assignment { dataout[5] } { LOCATION = M10; IOSTANDARD = LVCMOS33;  
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set_pin_assignment { dataout[6] } { LOCATION = D16; IOSTANDARD = LVCMOS33;  
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set_pin_assignment { dataout[7] } { LOCATION = E15; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { dataout[8] } { LOCATION = F14; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { dataout[9] } { LOCATION = F15; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { en[0] } { LOCATION = D14; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
= 8; PULLTYPE = NONE; }  
set_pin_assignment { en[10] } { LOCATION = M15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
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= 8; PULLTYPE = NONE; }
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set_pin_assignment { en[13] } { LOCATION = N14; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
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set_pin_assignment { en[4] } { LOCATION = H13; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
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set_pin_assignment { en[5] } { LOCATION = H14; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
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set_pin_assignment { en[8] } { LOCATION = K16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
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set_pin_assignment { rst } { LOCATION = T6; IOSTANDARD = LVCMOS33; PULLTYPE =  
PULLUP; }
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5、下载测试

将 16x16 点阵模块插在开发板 J1 口上。

下载程序之后。可以看到点阵循环显示“杭州康芯”。

